

## Unit 8 Congruence and Similarity (& their application)

Name: \_\_\_\_\_

Class: \_\_\_\_\_

### Unit Enduring Understandings

Students will be able to understand that

- Diagrams of figures help us to visualize their congruence or similarity.
- Two figures are congruent if and only if their sides and angles remain invariant under translation, rotation and reflection.
- Two similar figures have corresponding sides that are proportional.

### Unit Essential Questions

- How do diagrams facilitate problem solving?
- How do properties of congruent figures remain invariant under transformations?
- How does proportionality undergird the concept of similarity?

### Reference

- Textbook: *think! Mathematics New Syllabus Mathematics 2B (8<sup>th</sup> edition)* **Chapter 8**
- Textbook: *think! Mathematics New Syllabus Mathematics 3B (8<sup>th</sup> edition)* **Chapter 9 and 10**

### Knowledge & Skills (from O Level Syllabus)

| G2. Congruence and Similarity |                                                                                                                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1                           | Congruent figures                                                                                                                                                                |
| 2.2                           | Similar figures                                                                                                                                                                  |
| 2.3                           | Properties of similar triangles and polygons: <ul style="list-style-type: none"> <li>• Corresponding angles are equal</li> <li>• Corresponding sides are proportional</li> </ul> |
| 2.4                           | Enlargement and reduction of a plane figure                                                                                                                                      |
| 2.6                           | Solving simple problems involving congruence and similarity                                                                                                                      |
| 2.7                           | Determining whether two triangles are: <ul style="list-style-type: none"> <li>• Congruent</li> <li>• Similar</li> </ul>                                                          |
| 2.8                           | Ratio of areas of similar plane figures                                                                                                                                          |
| 2.9                           | Ratio of volumes of similar solids                                                                                                                                               |

## Teaching To The Big Idea ...

| Lesson sequence in the unit                                                        |                                                |                        |                       |                      |                      |                         |                             |                    |
|------------------------------------------------------------------------------------|------------------------------------------------|------------------------|-----------------------|----------------------|----------------------|-------------------------|-----------------------------|--------------------|
| Student Learning Outcomes                                                          | Dimensions (Please tick the appropriate boxes) |                        |                       |                      |                      |                         |                             |                    |
|                                                                                    | FUNCTIONS<br><b>F</b>                          | INVARIANCE<br><b>I</b> | NOTATIONS<br><b>N</b> | DIAGRAMS<br><b>D</b> | MEASURES<br><b>M</b> | EQUIVALENCE<br><b>E</b> | PROPORTIONALITY<br><b>P</b> | MODELS<br><b>M</b> |
| Congruent Figures, Tests for Congruent Triangles                                   |                                                | √                      |                       | √                    |                      |                         |                             |                    |
| Similarity, Enlargement and Reduction of Plane Figure, Tests for Similar Triangles |                                                |                        |                       | √                    |                      |                         | √                           |                    |
| Ratios and Volumes of Similar Figures                                              |                                                |                        |                       | √                    |                      |                         | √                           |                    |

### Big Idea on Proportionality

- The main theme of **Proportionality** are Properties and Relationships.
- **Proportionality** is a relationship between two quantities that allows one quantity to be computed from the other based on multiplicative reasoning.
- In **geometry**, proportional relationships undergird important concepts such as similarity and scales.
- Underlying the concept of proportionality are two quantities that vary in such a way that the ratio between them remains a constant.

### Checklist for Self-Assessment (Please tick ✓ at the appropriate boxes)

| Levels  | Characteristics                                                                                                                                                            | Knowledge & Skills                                                                                                     | No I really need help. I need to see an example | I still need some help. | Yes I can do this on my own. | Yes I can explain to others too. |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------|------------------------------|----------------------------------|
| Level 0 | Reproduction of facts, rules, formulae;<br>No explanations required                                                                                                        | State the definition of congruence and similarity.                                                                     |                                                 |                         |                              |                                  |
| Level 1 | Procedural tasks without connections,<br>Algorithmic in nature;<br>No explanations required                                                                                | Identify the similarity / differences between Similar Figures and Congruent Figures and find unknown sides and angles. |                                                 |                         |                              |                                  |
| Level 2 | Procedural tasks with connections,<br>Algorithmic in nature;<br>Has a “real-world” context;<br>Explanations required                                                       | Determine whether two or more triangles are congruent in a given context and able to show proof.                       |                                                 |                         |                              |                                  |
|         |                                                                                                                                                                            | Determine whether two or more triangles are similar in a given context and able to show proof.                         |                                                 |                         |                              |                                  |
| Level 3 | Problem Solving,<br>Non-algorithmic in nature;<br>Requires understanding and application of mathematical concepts;<br>Has a “real-world” context;<br>Explanations required | Solve problem in real world context using the relationship between similar figures and solids.                         |                                                 |                         |                              |                                  |

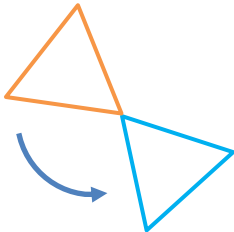
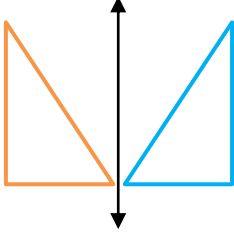
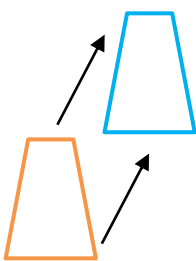
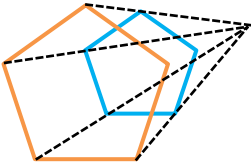
## Unit Key words

Ratio, proportion, corresponding, scale, enlargement, reduction, similarity, and congruence

Click this link to watch: [DIFFERENCE BETWEEN SIMILAR & CONGRUENT FIGURES?](#)

## Setting the Context: PRE-READINGS

Let us study the following types of Transformation to give you a better understanding on the concept of Congruency.

| A<br>Rotation                                                                                 | B<br>Reflection                                                                             | C<br>Translation                                                                    | D<br>Dilation /<br>Enlargement                                                      |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|              |            |  |  |
| A <b>ROTATION</b> turns the original figure about a point, called <b>CENTRE OF ROTATION</b> . | A <b>REFLECTION</b> flips the original figure about a <b>LINE</b> to create a mirror image. | A <b>TRANSLATION</b> slides the original figure about a line                        | A <b>DILATION</b> changes the size of the original figure.                          |

‘A, B and C are examples of **congruent** figures. While D is not.’

Write your observations on the lengths and angles of CONGRUENT figures from the examples above.

*Corresponding lengths and angles are equal.*

## What is congruence?

**Congruence** is a property of geometrical figures.

Two figures are **congruent** if they have exactly the **same shape and the same size**.

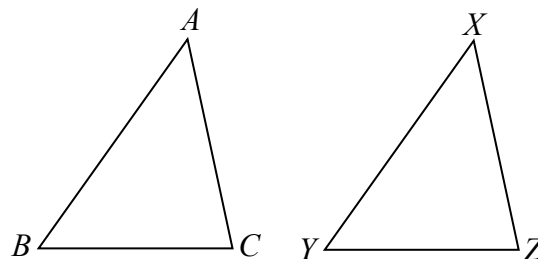
Two congruent figures can be **mapped onto each other under translation, rotation and reflection**.

## Labeling in Congruent Figures

Nomenclature in naming polygons, in this case **triangles**.

- Write a different letter at each vertex (corner) and list consecutive vertices in order.
- Two sides that have the same length are called corresponding congruent sides.

Given that triangle  $ABC$  and triangle  $XYZ$  are congruent, the following can be deduced:



Complete the following table.

| Corresponding vertices/points | Corresponding sides                     | Corresponding angles                   |
|-------------------------------|-----------------------------------------|----------------------------------------|
| $A$ corresponds to $X$        | $AB$ corresponds to $XY$<br>(not $YX$ ) | Angle $BAC$ corresponds to Angle $YXZ$ |
| $B$ corresponds to $Y$        | $BC$ corresponds to $YZ$                | Angle $ABC$ corresponds to Angle $XYZ$ |
| $C$ corresponds to $Z$        | $CA$ corresponds to $ZX$                | Angle $BCA$ corresponds to Angle $YZX$ |

Triangle  $ABC$  is congruent to triangle  $XYZ$  can also be written as:

$$\text{Triangle } ABC \equiv \text{Triangle } XYZ$$

This is called the statement of congruence, where the symbol ' $\equiv$ ' means 'is congruent to'.

Note: In the above example, writing "triangle  $ABC$  is congruent to triangle  $ZYX$ " is incorrect. Why?

→ Vertex A corresponds to vertex X .

Consider triangle  $ABC$  and triangle  $DEF$ , where

$$AB = DE$$

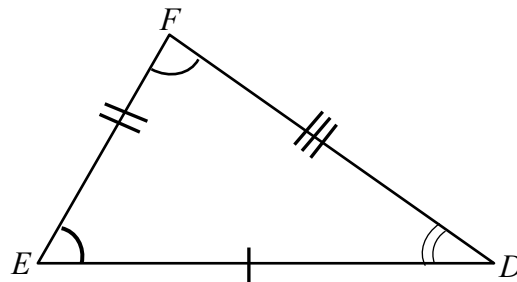
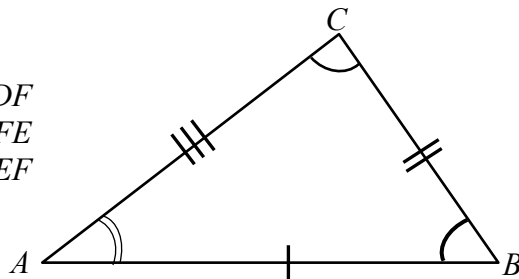
$$AC = DF$$

$$BC = EF$$

$$\text{Angle } BAC = \text{Angle } EDF$$

$$\text{Angle } ACB = \text{Angle } DFE$$

$$\text{Angle } ABC = \text{Angle } DEF$$



Triangle  $ABC$  is still congruent to triangle  $DEF$  even though triangle  $DEF$  is laterally inverted. This is because triangle  $ABC$  can be mapped onto triangle  $DEF$  by a reflection.

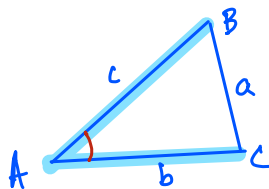
## Congruence Test - Discussion

Based on what you have observed,

1. What sets of conditions can we use to prove that two triangles are congruent?

*Answer: 3 corresponding sides are equal.*

2. What do you think is an 'included angle'? Illustrate with a diagram.

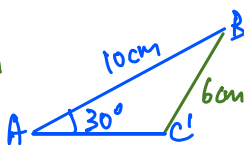
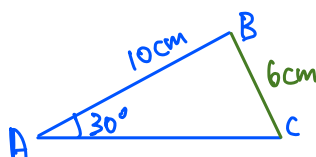


*With reference to AB and AC,  
 $\angle BAC$  is the included angle.*

*With reference to — and —  
 $\angle$  — is also an included angle*

- 3 Explain why the SSA test will NOT prove triangles are congruent.  
Why is SAS different from SSA?

*Side Side Angle*

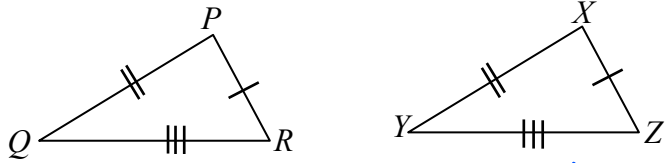
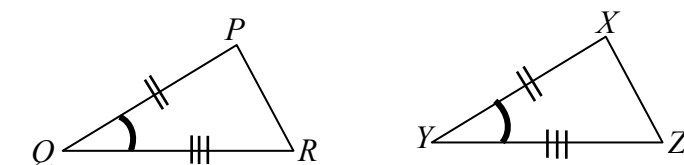
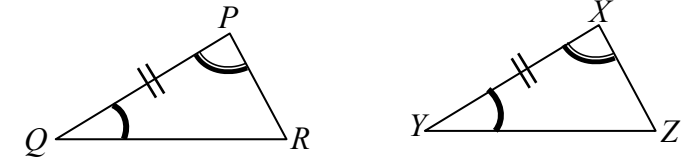
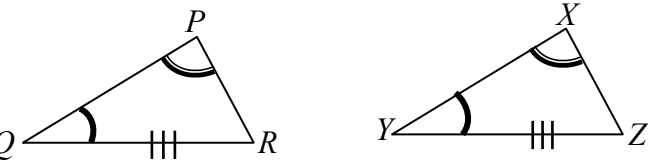
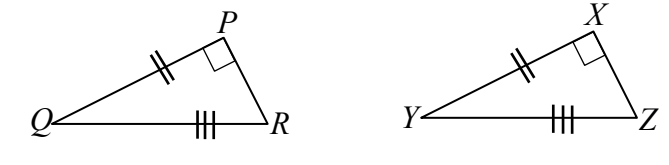


*With 2 sides and 1  
non-included angle,  
the two triangles may  
not be congruent.*

*So SSA cannot be used to triangles are congruent.*

Memorise the 4 tests and what to look for.

## Tests for Congruent Triangles (Summary)

|                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. <b>Side Side Side Congruency Test (SSS Congruence Test)</b></p> <p>If the <b>three sides</b> of one triangle are equal to the <b>three sides</b> of the other triangle, then the two triangles are congruent.</p>                                                                                      |  <p> <math>PQ = XY</math> (Side)<br/> <math>QR = YZ</math> (Side)<br/> <math>RP = ZX</math> (Side)<br/> <math>\triangle PQR \equiv \triangle XYZ</math>. (SSS Congruence Test) (Shown)         </p> <p>Memorise how to present.</p>           |
| <p>2. <b>Side Angle Side Test (SAS Congruence Test)</b></p> <p>If <b>two sides</b> and the <b>included angle</b> of one triangle are equal to <b>two sides</b> and the <b>included angle</b> of the other triangle, then the two triangles are congruent.</p>                                                |  <p> <math>PQ = XY</math> (Side)<br/> <math>\angle PQR = \angle XYZ</math> (Angle)<br/> <math>QR = YZ</math> (Side)<br/> <math>\triangle PQR \equiv \triangle XYZ</math>. (SAS Congruence Test) (Shown)         </p>                          |
| <p>3A. <b>Angle Side Angle Test (ASA Congruence Test)</b></p> <p>If the <b>two angles</b> and the <b>included side</b> of one triangle are equal to two angles and the included side of the other triangle, then the two triangles are congruent.</p>                                                        |  <p> <math>\angle PQR = \angle XYZ</math> (Angle)<br/> <math>QP = YX</math> (Side)<br/> <math>\angle QPR = \angle YXZ</math> (Angle)<br/> <math>\triangle PQR \equiv \triangle XYZ</math>. (ASA Congruence Test) (Shown)         </p>        |
| <p>3B. <b>Angle Angle Side Test (AAS Congruence Test)</b></p> <p>If <b>two angles</b> and <b>one side</b> (not included) of one triangle are equal to two angles and the corresponding side of the other triangle, then the two triangles are congruent.</p>                                                 |  <p> <math>\angle PQR = \angle XYZ</math> (Angle)<br/> <math>\angle QPR = \angle YXZ</math> (Angle)<br/> <math>QR = YZ</math> (Side)<br/> <math>\triangle PQR \equiv \triangle XYZ</math>. (AAS Congruence Test) (Shown)         </p>       |
| <p>4. <b>Right-angle Hypotenuse Side Test (RHS Congruence Test)</b></p> <p>If the <b>hypotenuse</b> (longest side) and <b>one side</b> of a <b>right-angled</b> triangle are equal to the hypotenuse and one side of the other right-angled triangle, then the two right-angled triangles are congruent.</p> |  <p> <math>\angle PQR = \angle XYZ = 90^\circ</math> (Right Angle)<br/> <math>QR = YZ</math> (Hypotenuse)<br/> <math>PQ = XY</math> (Side)<br/> <math>\triangle PQR \equiv \triangle XYZ</math>. (RHS Congruence Test) (Shown)         </p> |

Combine as AAS

① memorise : SSS, SAS, AAS, RHS

## Tests for Congruent Triangles (Summary)

For 2 congruent triangles,

- their corresponding angles are equal.
- their corresponding sides are equal.

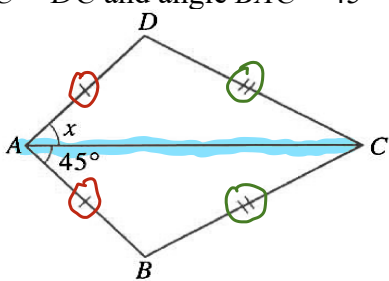
To prove that two triangles are congruent, include:

- 3 statements with reasons, vertices must correspond.
- use the symbol " $\equiv$ " to mean "is congruent to"
- conclusion with test used.

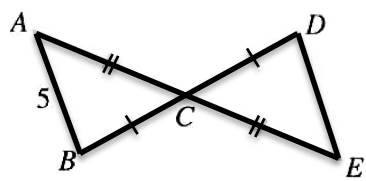
|| how to present.

## Tests for Congruent Triangles (Worked Examples)

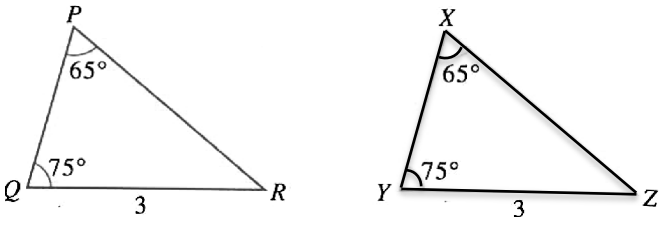
### (1) Congruence Test: SSS

|                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                           |                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <p><b>Question</b></p> <p>(1) SSS, SAS, AAS or RHS?</p>                                                                                   | <p>In the figure, <math>AB = AC</math>, <math>BC = DC</math> and angle <math>BAC = 45^\circ</math></p>  <p>Prove that triangle <math>ABC</math> and triangle <math>ADC</math> are congruent.</p>                                               |                                                                                                                                           |                                                                           |
| <p><b>Solution</b></p>                                                                                                                    | <table border="1"> <tr> <td> <math>AC = AC</math> (common side)<br/> <math>AB = AD</math> (given)<br/> <math>BC = DC</math> (given)<br/> <math>\triangle ABC \equiv \triangle ADC</math>. (SSS Congruence Test) (Shown) </td> <td> <b>Think:</b> To use SSS congruence test, look for 3 corresponding sides. </td> </tr> </table> | $AC = AC$ (common side)<br>$AB = AD$ (given)<br>$BC = DC$ (given)<br>$\triangle ABC \equiv \triangle ADC$ . (SSS Congruence Test) (Shown) | <b>Think:</b> To use SSS congruence test, look for 3 corresponding sides. |
| $AC = AC$ (common side)<br>$AB = AD$ (given)<br>$BC = DC$ (given)<br>$\triangle ABC \equiv \triangle ADC$ . (SSS Congruence Test) (Shown) | <b>Think:</b> To use SSS congruence test, look for 3 corresponding sides.                                                                                                                                                                                                                                                         |                                                                                                                                           |                                                                           |

### (2) Congruence Test: SAS

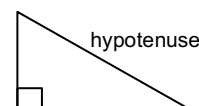
|                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                     |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <p><b>Question</b></p>                                                                                                                                              | <p>In the figure, line segments <math>AE</math> and <math>BD</math> bisect each other at <math>C</math>, and <math>AB = 5</math> cm.</p>  <p>Prove that triangle <math>ABC</math> and triangle <math>EDC</math> are congruent.</p>                                                                      |                                                                                                                                                                     |                                                                                                 |
| <p><b>Solution</b></p>                                                                                                                                              | <table border="1"> <tr> <td> <math>AC = EC</math> (given)<br/> <math>BC = DC</math> (given)<br/> <math>\angle ACB = \angle ECD</math> (vert. opp. <math>\angle</math>s)<br/> <math>\triangle ABC \equiv \triangle EDC</math>. (SAS Congruence Test) (Shown) </td> <td> <b>Think:</b> To use SAS congruence test, look for 2 corresponding sides and an included angle. </td> </tr> </table> | $AC = EC$ (given)<br>$BC = DC$ (given)<br>$\angle ACB = \angle ECD$ (vert. opp. $\angle$ s)<br>$\triangle ABC \equiv \triangle EDC$ . (SAS Congruence Test) (Shown) | <b>Think:</b> To use SAS congruence test, look for 2 corresponding sides and an included angle. |
| $AC = EC$ (given)<br>$BC = DC$ (given)<br>$\angle ACB = \angle ECD$ (vert. opp. $\angle$ s)<br>$\triangle ABC \equiv \triangle EDC$ . (SAS Congruence Test) (Shown) | <b>Think:</b> To use SAS congruence test, look for 2 corresponding sides and an included angle.                                                                                                                                                                                                                                                                                             |                                                                                                                                                                     |                                                                                                 |
| <p><b>Remark</b></p>                                                                                                                                                | <p>Vertices must correspond for all statements and conclusion.<br/> <i>i.e. "Triangle <math>ABC</math> is congruent to triangle <math>DEC</math>" is considered incorrect.</i></p>                                                                                                                                                                                                          |                                                                                                                                                                     |                                                                                                 |

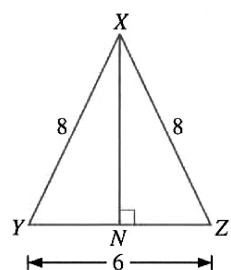
### (3) Congruence Test: ASA / AAS

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question |  <p>Prove that triangle <math>PQR</math> and triangle <math>XYZ</math> are congruent.</p>                                                                                                                                                                                                                                                                                                                      |
| Solution | <div style="display: flex; justify-content: space-between;"> <div style="width: 60%;"> <math>\angle PQR = \angle XYZ = 75^\circ</math> (given)<br/> <math>\angle RPQ = \angle ZXY = 65^\circ</math> (given)<br/> <math>QR = YZ = 3</math> units (given)<br/> <math>\triangle PQR \equiv \triangle XYZ</math>. (AAS Congruence Test) (Shown) </div> <div style="width: 35%;"> <b>Think:</b> To use ASA congruence test, look for <b>2 corresponding angles</b> and <b>one side</b>. </div> </div> |

### (4) Congruence Test: RHS

To use RHS Congruence Test, you need to know the hypotenuse. The **hypotenuse** is the name of the longest side of a **right-angled triangle**, located **opposite the right angle** (90 degrees).



|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question | <p>In the figure, <math>XY = XZ = 8</math> cm, <math>YZ = 6</math> cm and angle <math>XNZ = 90^\circ</math>. Prove that triangle <math>XYN</math> and triangle <math>XZN</math> are congruent.</p>                                                                                                                                                                                                                                             |
| Solution | <div style="display: flex; justify-content: space-between;"> <div style="width: 60%;"> <math>XY = XZ = 8</math> cm (given). Note: <b>hypotenuse</b><br/> <math>XN = XN</math> (common)<br/> <math>\angle XNY = \angle XNZ = 90^\circ</math> (given right-Angled)<br/> <math>\triangle XYN \equiv \triangle XZN</math>. (RHS Congruence Test) (Shown) </div> <div style="width: 35%;"> <b>Think:</b> To use RHS congruence test, look for corresponding <b>right-angle</b>, <b>hypotenuse</b> and <b>another side</b>. </div> </div> |

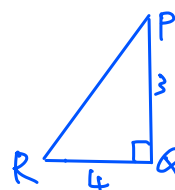
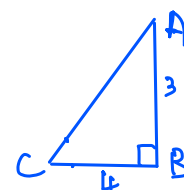
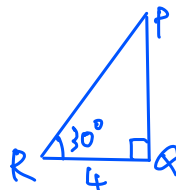
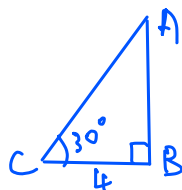
Click this link to watch: [Triangle Congruence: Does Angle-Side-Side Work?](#)

### Congruence Test - Discussion

Based on what you have observed, do we only use RHS congruence test for two right angled triangles?

Use examples to illustrate your points.

No. AAS can also work. or SAS



### Task 1 (guided)

Prove that triangle  $DEF$  and triangle  $PQR$  are congruent.

use this to write the corresponding sides/angles.

Step ①: Think: SSS, SAS, AAS or RHS?

$$DE = \frac{PQ}{PR} = \frac{61}{92} \text{ m} \quad (\text{Given}) [\text{Side}]$$

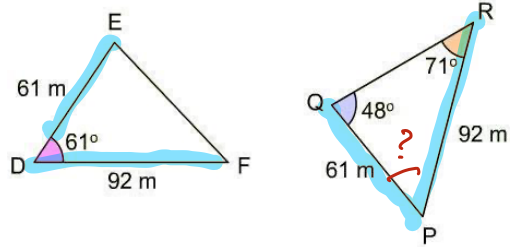
$$DF = \frac{PR}{} = \frac{92}{92} \text{ m} \quad (\text{Given}) [\text{Side}]$$

$$\angle QPR = \frac{180^\circ - 48^\circ - 71^\circ}{= 61^\circ} \quad (\text{Reasons: } \angle \text{sum of } \Delta)$$

$$\text{Thus, } \angle EDF = \angle QPR = 61^\circ \quad (\text{note: included angle})$$

$$\triangle DEF \equiv \triangle PQR \quad (\text{SAS Congruence Test}) \quad (\text{Shown})$$

- Included angle reference <https://www.mathsisfun.com/definitions/included-angle.html>



### Task 2 (fundamental)

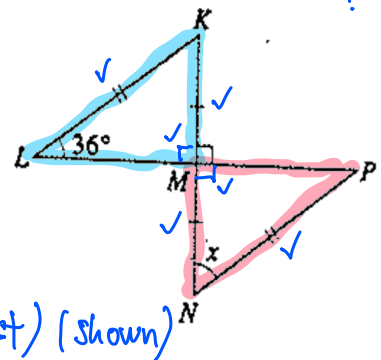
In the diagram below,  $KMN$  and  $LMP$  are straight lines. Angle  $KMP = 90^\circ$ .

- Prove that triangle  $KLM$  is congruent to triangle  $NPM$ . Think: SSS, SAS, AAS or RHS?
- Find the value of unknown angle  $x$ .

$$\begin{aligned} \therefore \angle LMK &= \angle PMN = 90^\circ \\ \angle K &= \angle N \quad (\text{Given}) \\ KM &= NM \quad (\text{Given}) \end{aligned}$$

$$\triangle KLM \equiv \triangle NPM \quad (\text{RHS Congruence Test}) \quad (\text{shown})$$

$$\begin{aligned} \text{(ii)} \quad \angle x &= \angle MNP \\ &= \angle MKL \\ &= 180^\circ - 90^\circ - 36^\circ \quad (\angle \text{sum of } \Delta) \\ &= 54^\circ \end{aligned}$$



### Task 3 (guided): Readiness

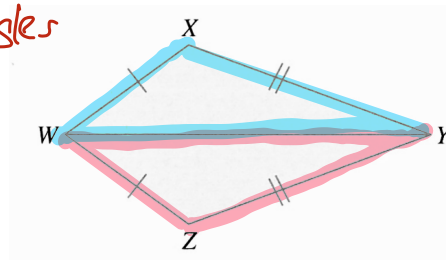
The diagram shows a kite  $WXYZ$ . It is given that  $WX = WZ$  and  $XY = ZY$ .

- Name 2 congruent triangles.
- Prove that the 2 triangles named in (i) are congruent.

**Solution** ① Identify the 2 congruent triangles

- |                    |          |           |
|--------------------|----------|-----------|
| $W$ corresponds to | <u>W</u> | Thinking! |
| $X$ corresponds to | <u>Z</u> |           |
| $Y$ corresponds to | <u>Y</u> |           |

②  $\triangle WXY$  is congruent to  $\triangle WZY$ .



- [must state test as it's a proving question]

$$WX = \underline{WZ} \quad (\text{given}) [S]$$

$$XY = \underline{ZY} \quad (\text{given}) [S]$$

$$WY = \underline{WY} \quad (\text{common}) [S]$$

$\triangle WXY$  is congruent to  $\triangle WZY$ . (SSS Congruence Test) (Shown)

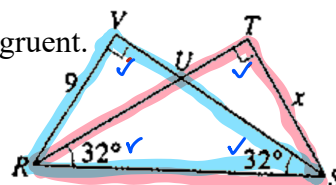
SSS?  
SAS?  
AAS?  
RHS?

### Task 4 (guided)

The diagram below shows two right-angled triangles  $RVS$  and  $RTS$ .

It is also given that angle  $TRS = \text{angle } VSR = 32^\circ$  and  $VR = 9$  m.

- Identify two congruent triangles and prove that they are congruent.
- Find the value of the unknown length  $x$ .



**Solution**

Complete the following. It could be side or angle.

- |          |                |          |
|----------|----------------|----------|
| <u>R</u> | corresponds to | <u>S</u> |
| <u>V</u> | corresponds to | <u>T</u> |
| <u>S</u> | corresponds to | <u>R</u> |

$$\angle RSV = \angle SRT = 32^\circ \text{ (given)}$$

$$\text{Angle } RVS = \text{Angle } STR = 90^\circ \text{ (given)}$$

$RS$  is a common side (hypotenuse)

Hence,  $\triangle RVS$  is congruent to  $\triangle STR$ . (AAS Congruence Test) (Shown)  
(Note that order of the vertices is important)

Think: SSS, SAS, AAS or RHS?  
SSS:  $\checkmark \times \times$ , SAS:  $\checkmark \times \times$ , AAS:  $\checkmark \checkmark \checkmark$ , RHS:  $\checkmark \checkmark \times$

$(\therefore) \triangle RVS$  and  $\triangle STR$

- From observation,  $TS = x = \underline{VR} = 9$  m.  
[This could only be done after you have established the two congruent triangles in (i)]

### Task 5 (consolidate) Higher Order Thinking Question

In the figure,  $\triangle ABC$  and  $\triangle CDE$  are equilateral triangles.  
Explain why  $AE$  is equal to  $BD$ .

(Hint: Establish the congruency of a pair of triangles, which consist of the sides  $AE$  and  $BD$ )

$\triangle ACE$  and  $\triangle BCD$

SSS, SAS, AAS, RHS ?

included  $\angle$

$$EC = BC \quad (\text{given})$$

$$CA = CB \quad (\text{given})$$

Let  $\angle ACD$  be  $x$

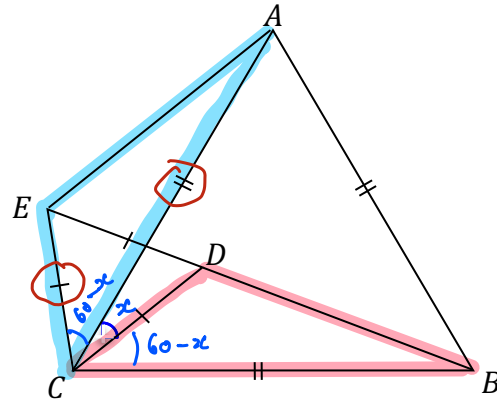
$$\therefore \angle ECA = 60^\circ - x \quad (\text{base } \angle\text{s, equilateral } \triangle ECD)$$

$$\angle DCB = 60^\circ - x \quad (\text{base } \angle\text{s, equilateral } \triangle ACB)$$

$$\therefore \angle ECA = \angle DCB$$

$$\therefore \triangle ACE \equiv \triangle BCD \quad (\text{SAS Congruence Test})$$

$$\therefore AE = BD$$



Prove the corresponding included angles are equal.

# Similarity



## Task 1

Discuss the following with your partner.

- Which, if any, of the following could be regarded as being of the **same shape** as a square boxing ring of side 5 m?
 

|                                                           |       |
|-----------------------------------------------------------|-------|
| (A) A square carpet measuring 6 m by 6 m                  | Y / N |
| (B) A square field with sides 280 m                       | Y / N |
| (C) One of the smallest squares of a sheet of graph paper | Y / N |
  
- Which, if any, of the following could be regarded as being of the same shape as a football pitch measuring 100 m by 60 m?
 

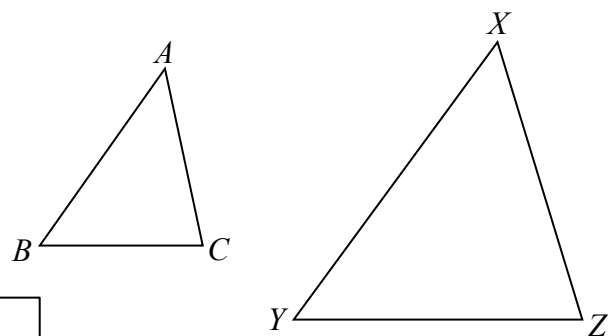
|                                                                           |       |
|---------------------------------------------------------------------------|-------|
| (A) A square 8 cm by 8 cm                                                 | Y / N |
| (B) A rectangle 5 m by 3 m                                                | Y / N |
| (C) A parallelogram with sides of 10 cm and 6 cm and one angle $85^\circ$ | Y / N |
| (D) A rectangle 10 m by 5 m                                               | Y / N |

Two polygons are **similar** if all the **corresponding angles** are **equal**, **AND** the **ratios of the corresponding sides** are **equal**.

**Note:** Congruence is a special case of similarity.

Given that triangle  $ABC$  and triangle  $XYZ$  are similar, the following can be deduced:

- Point  $A$  corresponds to point  $X$
- Point  $B$  corresponds to point  $Y$
- Point  $C$  corresponds to point  $Z$
  
- Angle  $BAC = \text{Angle } YXZ$
- Angle  $ABC = \text{Angle } XYZ$
- Angle  $BCA = \text{Angle } YZX$



- $\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ}$  Ratio of corresponding sides

When 2 triangles are similar, we will write it as “**is similar to**”  
(In the above example, writing “triangle  $ABC$  is similar to triangle  $ZYX$ ” is incorrect.)

### IMPORTANT:

- The order of the vertices of one triangle must correspond/ match to the vertices of the other triangle.

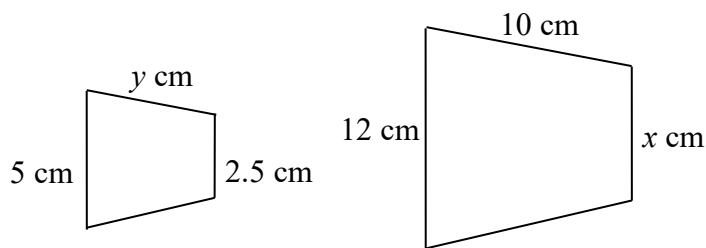
**2. There is no standard symbol for similarity. You MUST spell the word “similar”**

**Task 2**

The two figures shown below are similar.

The figure on the right is an enlargement of the figure on the left.

Find the values of  $x$  and  $y$ .



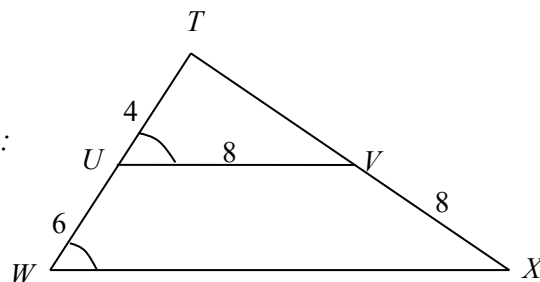
**Task 3**

Given that triangle  $UTV$  and triangle  $WTX$  are similar. Find the length of  $TV$  and  $WX$ .

Based on the similar triangles, write down the 3 pairs of ratios of corresponding sides as fractions:

$$\frac{UT}{WT} = \frac{TV}{WX} = \frac{UV}{WX}$$

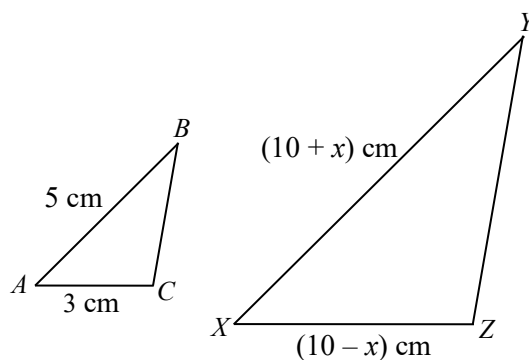
Then select the pairs that help you find the required lengths:



**Task 4**

Triangle  $ABC$  is similar to triangle  $XYZ$ . Given that  $AB = 5$  cm,  $AC = 3$  cm,

$XY = (10 + x)$  cm and  $XZ = (10 - x)$  cm, form an equation in  $x$  and solve it.





To prove that two triangles are similar, do we have to fulfill all corresponding angles are equal and all corresponding sides are proportional?

### To prove 2 triangles are similar

To **prove** that 2 triangles are similar, we need only prove that:

- all corresponding angles are equal.
- OR
- all corresponding sides are **proportional**.
- OR
- two pairs of corresponding sides are **proportional**, and the included angles are equal.

To **explain** that two triangles are similar, include:

- 3 statements with reasons
- conclusion with test used.

Note that the vertices of each triangle must correspond just like for congruence tests.

## Enlargement & Reduction of Plan Figures: Proportions

$$\text{Scale factor} = \frac{\text{Length of image}}{\text{Length of object}}$$

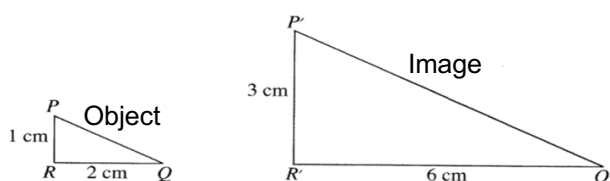
We multiply the distance between each point in an object by a scale factor to produce an image.

- The scale factor is greater than 1, the image produced is greater than the object
- The scale factor is between 0 and 1, the image produced is smaller than the object.

### Example

Triangle  $PQR$  is the object and triangle  $P'Q'R'$  is the image.

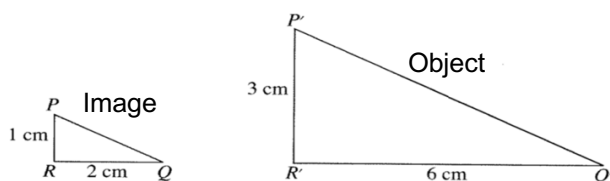
Triangle  $P'Q'R'$  is an enlargement of triangle  $PQR$  with a scale factor of 3.



$$\text{Scale factor} = \frac{P'R'}{PR} = \frac{R'Q'}{RQ} = 3$$

If we change the diagrams such that triangle  $PQR$  is the image and triangle  $P'Q'R'$  is the object.

Triangle  $PQR$  is an enlargement of triangle  $P'Q'R'$  with a scale factor of  $\frac{1}{3}$ .



$$\text{Scale factor} = \frac{PR}{P'R'} = \frac{RQ}{R'Q'} = \frac{1}{3}$$

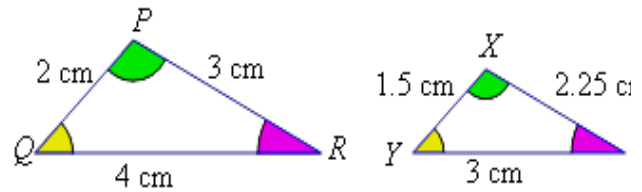
## Tests for Similar Triangles (With Explanations)

### (1) Similarity Test: SSS

$$\frac{PQ}{YX} = \frac{2}{1.5} = \frac{4}{3}$$

$$\frac{PR}{XZ} = \frac{3}{2.25} = \frac{4}{3}$$

$$\frac{QR}{YZ} = \frac{4}{3}$$

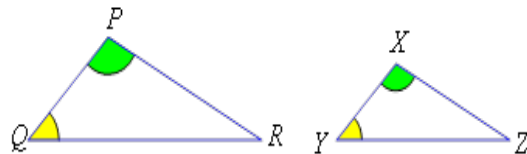


Since all the corresponding angles are equal and all the ratios of the lengths of the corresponding sides are equal, **triangle PQR is similar to triangle XYZ**. (SSS Similarity Test) (Shown)

### (2) Similarity Test: AA

Angle PQR = Angle XYZ (given)

Angle QPR = Angle YXZ (given)



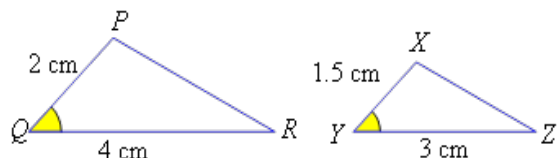
Since two corresponding angles of the two triangles are equal, **triangle PQR is similar to triangle XYZ**. (AA Similarity Test) (Shown)

**Essentially, we only need 2 corresponding pairs of angles. Why?**

### (3) Similarity Test: SAS

$$\frac{PQ}{YX} = \frac{2}{1.5} = \frac{4}{3}$$

$$\frac{QR}{YZ} = \frac{4}{3}$$



Angle PQR = Angle XYZ (given)

Since the ratios of two pairs of corresponding sides are equal and their **included** angles are equal, **triangle PQR is similar to triangle XYZ**. (SAS Similarity Test) (Shown)

**The proportion of the corresponding sides is known as the enlargement factor.**

## Similarity Tests (Worked Examples)

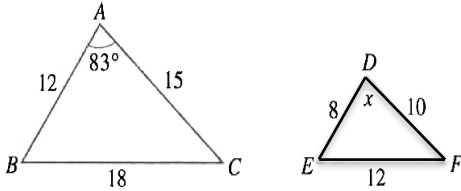
### (1) Similarity Test: AA

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question 1</b> | <p>In the figure, <math>AB</math> is parallel to <math>DE</math>, <math>BC = 8</math> cm, <math>CD = 6</math> cm and <math>CE = 5</math> cm.</p> <p>Prove that triangle <math>ABC</math> and triangle <math>EDC</math> are similar.</p> <div data-bbox="1125 336 1388 560"> </div>                                                                                                                                                                                                                        |
| <b>Solution</b>   | <div data-bbox="359 638 821 761"> <math>\angle BAC = \angle DEC</math> (alt. <math>\angle</math>s, <math>AB \parallel DE</math>)<br/> <math>\angle ABC = \angle EDC</math> (alt. <math>\angle</math>s, <math>AB \parallel DE</math>)<br/> <math>\angle ACB = \angle ECD</math> (vert. opp. <math>\angle</math>s) </div> <div data-bbox="1212 627 1404 761"> <p>Any 2 of these will do</p> </div> <p><math>\triangle ABC</math> is similar to <math>\triangle EDC</math>. (AA Similarity Test) (Shown)</p> |

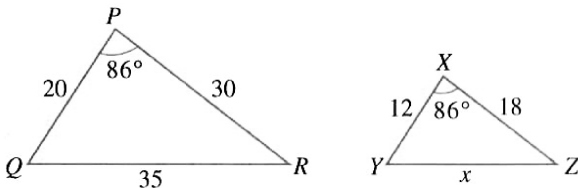
### (2) Similarity Test: AA

|                   |                                                                                                                                                                                                                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question 2</b> | <p><math>WXY</math> is a straight line and angle <math>WXZ =</math> angle <math>WZY</math>. It is also given that <math>WY = 15</math> cm, <math>WZ = 6</math> cm and <math>ZY = 4</math> cm.</p> <p>Prove that triangle <math>WYZ</math> and triangle <math>WZX</math> are similar.</p> <div data-bbox="829 1176 1372 1534"> </div> |
| <b>Solution</b>   | <div data-bbox="359 1612 798 1691"> <math>\angle WZY = \angle WXZ</math> (given)<br/> <math>\angle ZWY = \angle XWZ</math> (common angle) </div> <p><math>\triangle WYZ</math> is similar to <math>\triangle WZX</math>. (AA Similarity Test) (Shown)</p>                                                                            |

### (3) Similarity Test: SSS

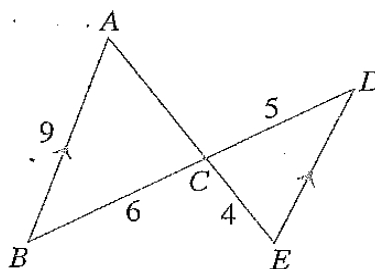
|                   |                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question 3</b> | <p>In the diagram, <math>AB = 12</math> cm, <math>BC = 18</math> cm, <math>CA = 15</math> cm, <math>DE = 8</math> cm, <math>EF = 12</math> cm, <math>FD = 10</math> cm and <math>\angle BAC = 83^\circ</math>. Prove that triangle <math>ABC</math> and triangle <math>DEF</math> are similar.</p>  |
| <b>Solution</b>   | $\frac{AB}{DE} = \frac{12}{8} = \frac{3}{2}$ $\frac{BC}{EF} = \frac{18}{12} = \frac{3}{2}$ $\frac{CA}{FD} = \frac{15}{10} = \frac{3}{2}$ $\therefore \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{3}{2}$ <p><math>\triangle ABC</math> is similar to <math>\triangle DEF</math>. (SSS Similarity Test) (Shown)</p>                                                           |

### (4) SAS Similarity Test

|                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question 4</b> | <p>In the diagram, <math>PQ = 20</math> cm, <math>QR = 35</math> cm, <math>RP = 30</math> cm, <math>XY = 12</math> cm, <math>ZX = 18</math> cm, <math>\angle RPQ = 86^\circ</math> and <math>\angle ZXY = 86^\circ</math>. Prove that triangle <math>PQR</math> and triangle <math>XYZ</math> are similar.</p>  |
| <b>Solution</b>   | <p><math>\angle RPQ = \angle ZXY = 86^\circ</math> (given)</p> $\frac{PQ}{XY} = \frac{20}{12} = \frac{5}{3}$ $\frac{RP}{ZX} = \frac{30}{18} = \frac{5}{3}$ $\therefore \frac{PQ}{XY} = \frac{RP}{ZX} = \frac{5}{3}$ <p><math>\triangle PQR</math> is similar to <math>\triangle XYZ</math>. (SAS Similarity Test) (Shown)</p>                                                                       |

## Task 5

In the diagram,  $AB = 9$  cm,  $BC = 6$  cm,  $CD = 5$  cm and  $CE = 4$  cm.  
It is also given that  $BA$  is parallel to  $ED$ .



- (a) Name two triangles that are similar.

**Think:**

$A$  corresponds to \_\_\_\_\_

$B$  corresponds to \_\_\_\_\_

$C$  corresponds to \_\_\_\_\_

*Solution:*  $\triangle ABC$  is similar to  $\triangle$  \_\_\_\_\_ ( \_\_\_\_\_ Similarity Test) (shown)

- (b) Calculate the lengths of  $AC$  and of  $DE$ .

*Solution:*

**Think:** Based on the similar triangles, write the 3 pairs of ratios of corresponding sides as fractions.

$$\frac{AB}{EC} = \frac{BC}{DC} = \frac{AC}{DE} = \text{---}$$

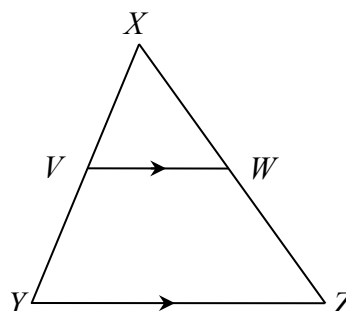
Then select the pairs that help you find the required lengths.

## Task 6

In the diagram,  $VX = 12$  cm,  $VW = 9$  cm and  $YZ = 15$  cm.

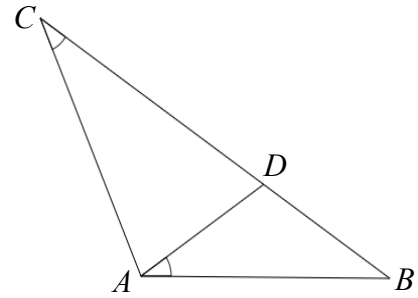
- (a) Show that triangle  $XVW$  is similar to triangle  $XYZ$ .  
(b) Calculate the value of  $VY$ .

**Think:** AA, SSS, SAS?



### Task 7

In the diagram,  $ABC$  is a triangle.  
 $D$  is the point on  $BC$  such that  $\angle BCA = \angle BAD$ .  
Given that  $BD = 4$  m and  $DC = 5$  m,  
find the length of  $AB$ .



**Think:**

1. Identify the two similar triangles.
2. Show that the two triangles are similar.
3. Use the ratio of the corresponding sides to find  $AB$ .

## Areas of Similar Figures

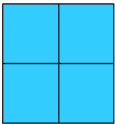
Consider a square A of side 2 cm. Notice what happens to the area when you double and treble its sides.

**A**



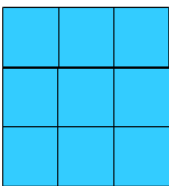
Area = 4 cm<sup>2</sup>

**B**



Area = 16 cm<sup>2</sup>

**C**



Area = 36 cm<sup>2</sup>

Ratio of the sides of squares A and B =  $\frac{2}{4}$

Ratio of the areas of squares A and B =  $\frac{4}{16}$

$$\frac{4}{16} = \left(\frac{2}{4}\right)^2$$

Ratio of the sides of squares A and C =  $\frac{2}{6}$

Ratio of the areas of squares A and C =  $\frac{4}{36}$

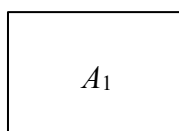
$$\frac{4}{36} = \left(\frac{2}{6}\right)^2$$

Ratio of the areas of A and B = (Ratio of their sides)<sup>2</sup>

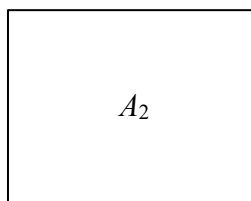
Ratio of the areas of A and C = (Ratio of their sides)<sup>2</sup>

When two figures are **similar**, the ratio of their areas is equal to the \_\_\_\_\_ of the ratio of any two \_\_\_\_\_ of the two figures.

If  $A_1$  and  $A_2$  denote the areas of **similar figures**, and  $l_1$  and  $l_2$  denote their corresponding lengths, we have



dimensions  
 $l_1$  by  $w_1$



dimensions  
 $l_2$  by  $w_2$

$$\frac{A_1}{A_2} = \frac{l_1 \times w_1}{l_2 \times w_2}$$

since

$$\frac{l_1}{l_2} = \frac{w_1}{w_2}$$

then

$$\frac{A_1}{A_2} = \frac{l_1 \times l_1}{l_2 \times l_2} = \left(\frac{l_1}{l_2}\right)^2$$

## Volumes of Similar Figures

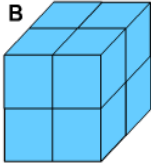
Use the same concept to derive the relationship between Volumes of similar figures.

Consider a solid A with side 2 cm. Notice what happens to the volume when you double and treble its sides



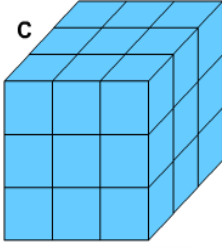
**A**

Volume =  $8 \text{ cm}^3$



**B**

Volume =  $64 \text{ cm}^3$



**C**

Volume =  $216 \text{ cm}^3$

Ratio of the sides of A and B =  $\frac{2}{4}$

Ratio of the volumes of A and B =  $\frac{8}{64}$

$$\frac{8}{64} = \left(\frac{2}{4}\right)^3$$

Ratio of the sides of A and C =  $\frac{2}{6}$

Ratio of the volumes of A and C =  $\frac{8}{216}$

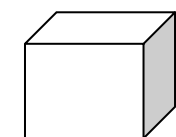
$$\frac{8}{216} = \left(\frac{2}{6}\right)^3$$

Ratio of the volumes of A and B = (Ratio of their sides)<sup>3</sup>

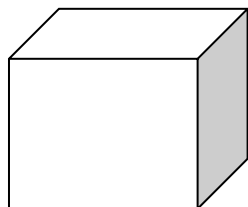
Ratio of the volumes of A and C = (Ratio of their sides)<sup>3</sup>

When two figures are **similar**, the ratio of their volumes is equal to the \_\_\_\_\_ of the ratio of any two \_\_\_\_\_ of the two figures.

If  $V_1$  and  $V_2$  denote the areas of **similar figures**, and  $l_1$  and  $l_2$  denote their corresponding lengths, we have



dimensions  
 $l_1$  by  $w_1$  by  $h_1$



dimensions  
 $l_2$  by  $w_2$  by  $h_2$

$$\frac{V_1}{V_2} = \frac{l_1 \times w_1 \times h_1}{l_2 \times w_2 \times h_2}$$

since

$$\frac{l_1}{l_2} = \frac{w_1}{w_2} = \frac{h_1}{h_2}$$

then

$$\frac{V_1}{V_2} = \frac{l_1 \times l_1 \times l_1}{l_2 \times l_2 \times l_2} = \left(\frac{l_1}{l_2}\right)^3$$

How to test that 2 **circles** are similar?

If they are similar, what would be the ratio of their surface area?

How to test that 2 **hemispheres** are similar?



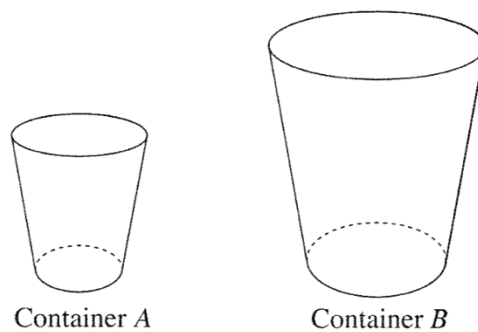
If they are similar what would be the ratio of their (i) surface area and (ii) volume?

### Summary: Areas and Volumes of Similar Figures

For two geometrically similar solids,

- $\frac{\text{length}_1}{\text{length}_2} = \frac{\text{height}_1}{\text{height}_2}$
- $\frac{\text{area}_1}{\text{area}_2} = \left(\frac{\text{length}_1}{\text{length}_2}\right)^2 = \left(\frac{\text{height}_1}{\text{height}_2}\right)^2$
- $\frac{\text{volume}_1}{\text{volume}_2} = \left(\frac{\text{length}_1}{\text{length}_2}\right)^3 = \left(\frac{\text{height}_1}{\text{height}_2}\right)^3$
- $\frac{\text{mass}_1}{\text{mass}_2} = \frac{\text{volume}_1}{\text{volume}_2} = \left(\frac{\text{length}_1}{\text{length}_2}\right)^3 = \left(\frac{\text{height}_1}{\text{height}_2}\right)^3$  if the two solids have same density.

### Volumes of Similar Figures (Worked Examples)



The diagram shows two geometrically similar containers  $A$  and  $B$ . The base areas of the containers  $A$  and  $B$  are  $16 \text{ cm}^2$  and  $25 \text{ cm}^2$  respectively.

- (a) Find the ratio of the heights of container  $A$  and container  $B$ .
- (b) Container  $A$  and  $B$  are filled with flour. The mass of flour in container  $B$  is  $7.5 \text{ kg}$ . Find the mass of flour in container  $A$ .

(a)

$$\frac{h_A}{h_B} = \sqrt{\frac{16}{25}}$$

$$\frac{h_A}{h_B} = \frac{4}{5}$$

Height of container  $A$  : Height of container  $B$   
 $= 4 : 5$

(b)

$$\frac{m_A}{m_B} = \left(\frac{4}{5}\right)^3$$

$$\frac{m_A}{7.5} = \frac{64}{125}$$

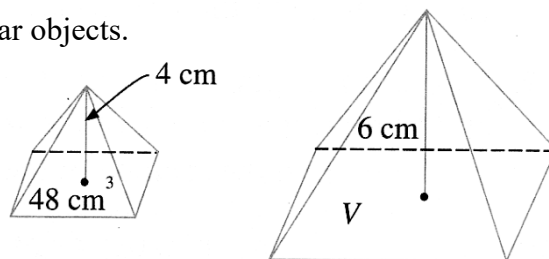
$$m_A = \frac{64}{125} \times 7.5$$

$$m_A = 3.84$$

mass of flour in container  $A$  is  $3.84 \text{ kg}$

### Task 9

Find the unknown volume  $V$  the following pairs of similar objects.



### Task 10

Two **similar** cones of the same material have base diameters 24 cm and 16 cm respectively.

The volume of the larger cone is  $378 \text{ cm}^3$  and the mass of the smaller cone 928 g.

(a) What is the significance of the terms 'same material'?

Calculate

(b) the volume of the smaller cone,

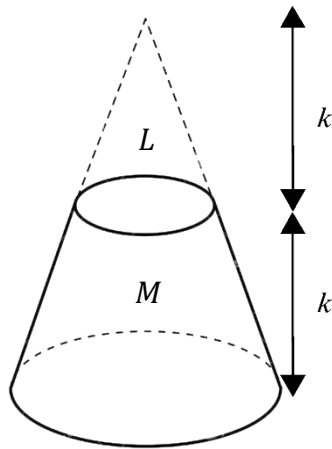
(c) the mass of the larger cone.

### Task 11

A right circular cone is divided into 2 portions,  $L$  and  $M$ , by a plane parallel to the base as shown in the diagram.

The height of each portion is  $k$  units.

Find the ratio of the volume of  $L$  to the volume of  $M$ .

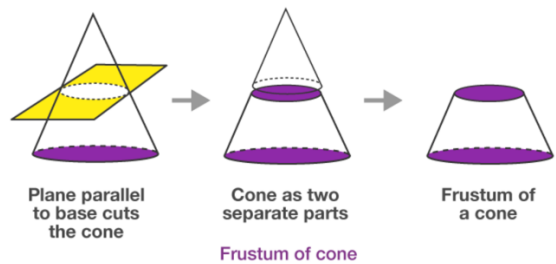


**Think:**

$L$  is not similar to  $M$ .

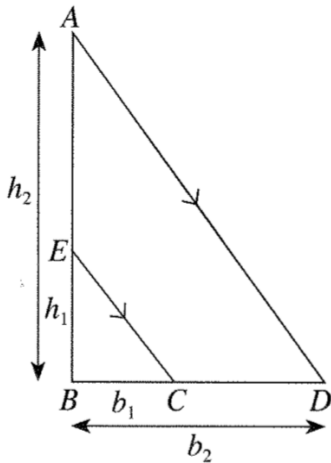
Yet they are part of a whole cone. How to use the concept of similarity to solve the problem.

$L$  is similar to \_\_\_\_\_.



## Finding Areas of Triangles Using Ratios

### (I) Similar Triangles



Consider two similar triangles  $\Delta BCE$  and  $\Delta BDA$

$$\frac{\text{Area of } \Delta BCE}{\text{Area of } \Delta BDA} = \left(\frac{b_1}{b_2}\right)^2$$

OR

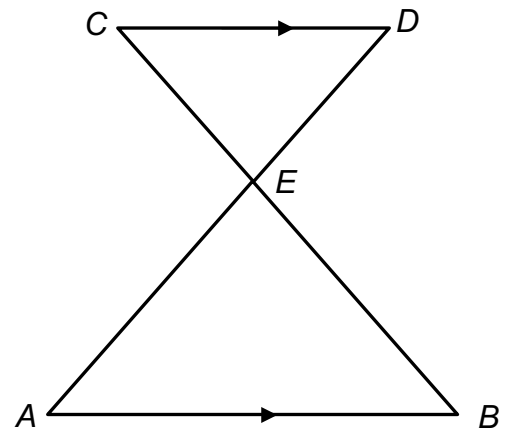
$$\frac{\text{Area of } \Delta BCE}{\text{Area of } \Delta BDA} = \left(\frac{h_1}{h_2}\right)^2$$

### Task 12

In the diagram,  $AED$  and  $BEC$  are straight lines.  
It is given that  $CD$  is parallel to  $AB$  and  $CD = \frac{2}{3}AB$ .

Find the ratio  $\frac{\text{area of } \Delta ABE}{\text{area of } \Delta DCE}$

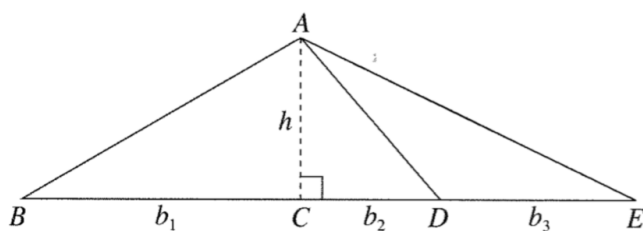
Solution:



Click this link to watch the video: [Ratio Of Areas Of Non-Similar and Non-Congruent Triangles](https://tinyurl.com/cp6svn9k)  
<https://tinyurl.com/cp6svn9k>

## Triangles Sharing Same Base / Height

### Ratios of Areas of Triangles with Common Heights

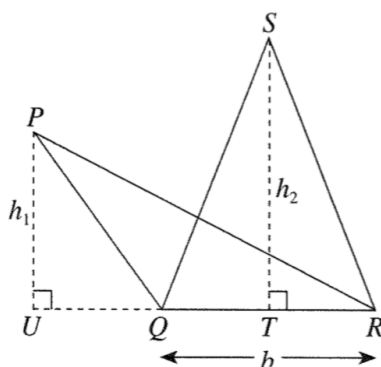


Consider  $\triangle ABC$ ,  $\triangle ACD$ ,  $\triangle ADE$ ,  $\triangle ABD$ ,  $\triangle ACE$  and  $\triangle ABE$  with common height,  $h$ .

$$\frac{\text{Area of } \triangle ACD}{\text{Area of } \triangle ADE} = \frac{\frac{1}{2} \times b_2 \times h}{\frac{1}{2} \times b_3 \times h} = \frac{b_2}{b_3}$$

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle ACE} = \frac{\frac{1}{2} \times b_1 \times h}{\frac{1}{2} \times (b_2 + b_3) \times h} = \frac{b_1}{b_2 + b_3}$$

### Ratios of Triangles with Common Base



Consider  $\triangle PQR$  and  $\triangle SQR$  with common base,  $b$ .

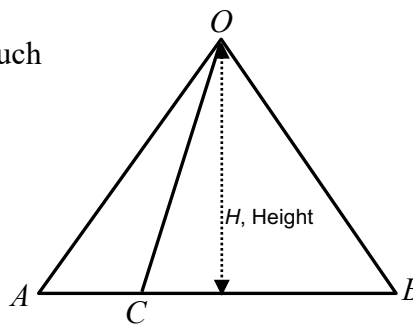
$$\frac{\text{Area of } \triangle PQR}{\text{Area of } \triangle SQR} = \frac{\frac{1}{2} \times h_1 \times b}{\frac{1}{2} \times h_2 \times b} = \frac{h_1}{h_2}$$

## Task 13

In the diagram,  $OAB$  is a triangle.  $C$  is a point on  $AB$  such that  $AC = \frac{2}{5}AB$ .

Find the ratio  $\frac{\text{area of } \triangle OAC}{\text{area of } \triangle OBC}$ .

Solution:



### Think:

- 1: Notice that triangle  $AOC$ , triangle  $OBC$  and triangle  $OAB$  are **NOT** congruent nor similar.
- 2: Triangle  $AOC$ , triangle  $OBC$  and triangle  $OAB$  have a common height.

Let  $h$  be the perpendicular height.

Use: Area of triangle =  $\frac{1}{2} \times \text{base} \times \text{height}$  to find the area of triangle  $OAC$  and triangle  $OBC$  using their respective base and height.

### Task 14 Higher Order Thinking Question

In the diagram,  $ABC$ ,  $AFD$  and  $CDE$  are straight lines and  $BF$  is parallel to  $CD$ .

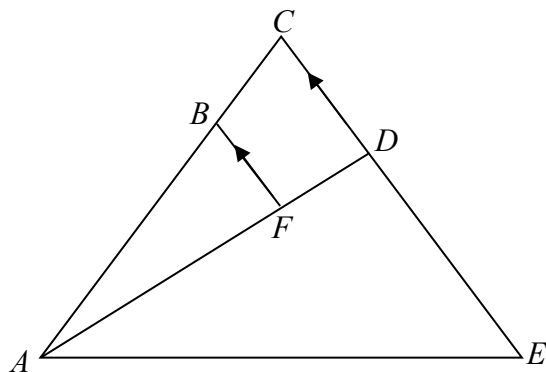
(a) **Show** that triangle  $ABF$  and triangle  $ACD$  are similar.

(b) Given that  $CD = \frac{1}{3} CE$  and  $BC = \frac{1}{2} AB$ . Find

(i)  $\frac{\text{Area of triangle } ABF}{\text{Area of triangle } ACD}$

(ii)  $\frac{\text{Area of triangle } ABF}{\text{Area of triangle } ADE}$  (Challenging)

Solution



END